

Cybersecurity Incident Report

SYN Flood (DoS) Attack Analysis

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Tool used: Wireshark (TCP/HTTP packet log)

Affected system: Company travel agency web server (192.0.2.1), sales webpage

Scenario

Employees and customers reported a connection timeout error when trying to access the company's sales webpage. An automated monitoring alert had already flagged a problem with the web server. As the analyst, TCP/HTTP traffic to and from the server was captured and analyzed using Wireshark to identify the cause.

Section 1: Identify the Type of Attack

One potential explanation for the connection timeout error:

the web server is overwhelmed by a flood of incomplete TCP connection requests, leaving it unable to respond to legitimate traffic in time.

What the logs show:

the source IP 203.0.113.0 sent 139 SYN packets to the server (192.0.2.1) on port 443, but never once completed the handshake by sending the final ACK. The server attempted to respond with 2 SYN-ACK packets and eventually sent 8 RST,ACK packets to forcibly close several of the abandoned half-open connections — but the flood of new SYN packets continued regardless.

Conclusion — this event could be:

a Denial of Service (DoS) attack — specifically a SYN flood — originating from a single source IP address (203.0.113.0) rather than a distributed network of attackers. Because the flood comes from one IP, this is classified as DoS rather than DDoS.

Section 2: How the Attack Caused the Website to Malfunction

The three steps of a normal TCP handshake:

1. The client sends a SYN packet to the server, requesting to start a connection.
2. The server responds with a SYN-ACK packet, acknowledging the request.
3. The client sends a final ACK packet, confirming the connection is established.

What happens when an attacker sends a large number of SYN packets at once:

The server reserves a slot in memory for each SYN it receives, holding the connection open while waiting for the final ACK. Because the attacker never sends that ACK, each connection stays half-open. The server has only a limited number of slots for pending connections, so as the attacker keeps sending more SYNs, those slots fill up faster than they can expire — eventually leaving no room for genuine connection attempts.

What the logs indicate and how that affects the server:

203.0.113.0 repeatedly sent SYN packets to port 443 without ever completing the handshake, while the server attempted to keep up via SYN-ACK replies and eventual RST,ACK resets. Meanwhile, a legitimate

request from 198.51.100.5 received a 504 Gateway Time-out, because server resources were consumed managing the flood of fake half-open connections instead of processing real visitor requests. This is what caused the website outage and connection timeout error reported in the original alert.

Consequences for the Organization

- Employees and customers were unable to access the sales webpage, directly disrupting business operations.
- Potential lost sales and customer trust due to the visible outage.
- IT team resources diverted to incident response rather than other planned work.

Recommended Mitigation Steps

- SYN cookies — avoid reserving memory until the handshake is cryptographically verified.
- Rate limiting — restrict SYN packets accepted per source IP within a time window.
- Stateful firewall / IDS tuning — detect abnormal patterns of incomplete handshakes from one source.
- IP blocking is a short-term measure only — attackers can spoof or rotate IPs to bypass a simple block.

Tools & Concepts Applied

- TCP three-way handshake (SYN, SYN-ACK, ACK)
- Packet capture and analysis with Wireshark
- DoS vs. DDoS classification based on source IP count
- Server resource exhaustion as an attack mechanism
- Incident root cause analysis and mitigation planning